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EXAMINING EVENT STUDY METHODOLOGIES IN FOREIGN EXCHANGE MARKETS

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Abstract. A Brown and Warner [1980, 1985] event study methodology is applied to the foreign exchange area. Comparisons of the performance of four abnormal return models are examined with simulations under different experimental conditions, such as choice of foreign currency or numeraire, level of abnormal shock, sample size, length of estimation period, market return proxy, and time period examined. The results provide practical suggestions on the selection of an event study methodology and demonstrate that some of the findings of Brown and Warner are not generalizable to the foreign exchange area.

Event studies examining the impact of particular types of events on the returns within a given financial market are mostly performed on common stocks. This serves as a major justification for Brown and Warner [1980, 1985] to evaluate event study methodologies with common stocks. Their analysis provides many useful suggestions in performing event studies that are now commonly adopted by others. The event study methodology can also be applied to other markets such as preferred stocks, bonds, options, commodities and currencies. Information on the generalizability of Brown and Warner's original findings can be obtained by performing their type of testing in some of these markets.

In order to both encourage and facilitate an expansion of event studies in other financial markets and provide methodological suggestions to future researchers, this study provides a modest first step by applying the Brown and Warner approach to the currency markets. If consistency is found between both their and our findings, the implications for acceptable methodologies are more likely generalizable over a wider spectrum of different types of security and non-security returns series. Alternatively, if material

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differences in results are found, caution will be needed in adopting a specific methodology based on tests of either common stock returns or currency market returns if anything other than common stocks or currencies are being subjected to event study analysis. The potential for either false acceptances or rejections of tests for abnormal returns may then not be determinable until a Brown and Warner-type simulation is performed for the specific type of security (or non-security) return series.

Market efficiency tests of weak and semi-strong forms have been conducted in the foreign exchange area. Among the semi-strong form tests, studies such as Frenkel [1981], Ito and Roley [1987] and Hardouvelis [1988] use the multiple regression approach employing changes in exchange rates as the dependent variable in testing the effects of public announcements on foreign exchange markets. On the other hand, Cosset and Doutriaux de la Rianderie [1985] and Sheffrin and Russell [1984] use the event study methodology as an alternative approach in examining these effects. The results of the present study will be useful to foreign exchange market efficiency tests which use the event study approach.

The discussion of the paper is divided into six major sections. After this introductory section, the second section reviews the literature on event studies while the third section describes the experimental design. The fourth section examines the properties of excess returns on event dates. Various simulations are then conducted and reported in the fifth section. The findings are then summarized in the last section.

LITERATURE REVIEW

Through simulation procedures, Brown and Warner [1980] examine various event study methodologies. Their objective is to determine the power of various event study techniques to correctly classify both the occurrence and nonoccurrence of abnormal events with common stock returns under different environmental conditions. Their 1980 study examines event methodologies using monthly security returns. In Brown and Warner [1985] the authors extend their evaluation of event study methodologies to daily common stock returns. They examine three return estimation models including the mean-adjusted return model, a market-adjusted return model, and a market model that employs an OLS procedure. The results show that simpler models are rather robust in correctly classifying the occurrence, or nonoccurrence, of abnormal returns. Brown and Warner [1980, 1985] also examine the sensitivity of the three models to a host of other environmental conditions. For example, they examine the impact of the size of the abnormal return, the certainty of the exact date of the abnormal return, change in variance of returns where there is an abnormal event, and changes in sample size. The current paper also examines many of the same conditions examined by Brown and Warner in addition to specific conditions that are found in the foreign exchange market.

Thus far, there are few applications of the event study methodology in foreign exchange markets. Two notable examples are Sheffrin and Russell [1984] and Cosset and Doutriaux de la Rianderie [1985]. Sheffrin and Russell analyzes the impact of North Sea oil discoveries on the value of sterling by looking at foreign exchange market reactions to announcements of oil discoveries. The authors find no evidence for the common view that oil discoveries led to an appreciation of sterling. Alternatively, Cosset and Doutriaux de la Rianderie examine the reaction of the currency market to the announcement of changes in the business environment of a country. The results show that news of political risk conveys important information about a country's investment climate and causes its currency's exchange rate to vary. Furthermore, the foreign exchange market is efficient in interpreting and reacting to this type of event. Besides these two studies, event studies in the foreign exchange market may also be applied to other interesting issues such as the announcement effects of money supply or various balance-of-payments account balances (e.g., trade account, current account, basic and overall balances).

EXPERIMENTAL DESIGN

Data Sources

In the calculation of abnormal returns in foreign exchange markets, exchange rate and interest-rate data are employed. The data used in this study span daily observations from January 1978 through February 1988, and are obtained from the Federal Reserve Bank. Complete information on daily interest rates is only available from 1978 for eight commonly traded currencies—the U.S. dollar, the Belgian franc, the Canadian dollar, the deutsche mark, the French franc, the Japanese yen, the Netherlands guilder, and the British pound. The scope of this study, therefore, is limited to these eight currencies and the ten-year and two-month period, providing a total of 2,650 business days of data.¹

Sample Construction

Following Brown and Warner [1980, 1985], multiple samples are used to test the power of different return-generating models. In the standard test to be performed, there are 100 samples where each sample consists of 50 independent draws of a consecutive 61-day time-series of exchange rates and interest rates for a given numeraire (reference currency) and a given foreign currency. The following process is used to select each of the 100 samples. First, for each sample, 50 days are randomly selected with replacement from the possible 3,650 days of available data. The randomly selected day is rejected and another is drawn if there is a missing exchange rate between the numeraire and the foreign currency or the daily interest on either currency is not available for day -1 or 0 . The eleven-day period $(-5, +5)$ is designated the event period and day 0 is the hypothetical event date used to test arbitrary shocks. Additionally, when the abnormal returns

are examined over the entire period $(-5, +5)$, no missing data on exchange or interest rates are allowed for the period $(-6, +5)$.

A critical date is also rejected and a new one drawn if there are less than 30 days of available information on both the exchange rate and two daily interest rates (numeraire and foreign currency) in the 50-day time-frame covering days -55 through -6 . The 50-day period is used to estimate parameters used in both the mean adjusted model and market model, which are used to generate abnormal returns models described below.

The above process is repeated 100 times to provide 100 samples, each containing 50 draws. The entire process is replicated for each new foreign currency. Thus, for a given numeraire, there are seven tests since the process is repeated for each of the seven foreign currencies. Treating each of the eight currencies as a numeraire, a total of 56 tests (8 numeraires with 7 foreign currencies each) are performed.

A set of 56 tests comprises a module that is used to examine a specific set of environmental conditions. For example, three modules, each containing 56 tests, are used in Simulation 2.1 to examine the sensitivity of test results to varying the level of abnormal shock induced on day 0.

Excess Return Measures

To measure abnormal returns, first a definition of returns in the foreign exchange market is needed. The return measure to be employed may be interpreted as the nominal return on speculation in the money market or in the foreign exchange market when interest-rate parity prevails. Let $R_{j,t}$ designate the estimate of the arithmetic daily return in the foreign exchange market of foreign currency j for time t where

$$\tilde{R}_{j,t} = \frac{E(\tilde{S}_{j,t} - S_{j,t-1})}{S_{j,t-1}} - (r_{n,t-1} - r_{j,t-1}), \quad (1)$$

where $E(.)$ is proxied by using the observed value of $S_{j,t} - S_{j,t-1}$, and $S_{j,t}$ is the spot exchange rate of currency j with respect to the numeraire currency n (in terms of number of units of the numeraire currency n per unit of the foreign currency j) at time t . The interest rates at time $t-1$ are $r_{n,t-1}$ and $r_{j,t-1}$ for currencies n and j , respectively. The interest rates are needed in the equation to take into account the interest differential between the two countries.²

Next, $A_{j,t}$ is defined as the excess return from currency j at day t . For each foreign currency, the excess return is estimated with four models for the critical day, or for each day in the event period $(-5, +5)$ when the event period is being examined. The random walk model is often mentioned in the foreign exchange market while the mean-adjusted, market-adjusted and market models are those discussed in Brown and Warner [1980, 1985].

1. Random walk model³

According to the Theory of the International Fisher Effect (also known as the Fisher Open Effect), the spot exchange rate should change in an equal but opposite direction to the difference in interest rates between two countries. In other words, the currency return is a martingale with expected value zero, $A_{j,t}=0$. Since the model is often mentioned in the foreign exchange market, its performance relative to the other models is examined in the present study.⁴

2. Mean-adjusted returns

$$A_{j,t} = \tilde{R}_{j,t} - \bar{R}_j, \quad (2)$$

$$\bar{R}_j = 1/50 \sum_{t=-6}^{-55} \tilde{R}_{j,t}, \quad (3)$$

where $\tilde{R}_j$ is the mean return of currency j in the estimation period, related to the specific critical day selected for a given draw of a 61-day time-series of returns. Equation (3) and subsequent equations are modified for missing observations by excluding days where $R_{j,t}$ are unavailable and modifying calculations to reflect the smaller number of available observations. A minimum of thirty observations are required in the estimation period.

3. Market-adjusted returns

$$A_{j,t} = \tilde{R}_{j,t} - \tilde{R}_{m,t}, \quad (4)$$

where $\tilde{R}_{m,t}$ is the return on the equally weighted portfolio of six foreign currencies. The numeraire and foreign currency j are excluded from the market return.⁵ Other proxies for market portfolio are covered in Simulation 5.

4. Market model

The Roll and Solnik [1977] foreign exchange asset pricing model is adopted as the market model in this study. Alternative models considered, but excluded in the testing, include Grauer, Litzenberger and Stehle [1976], Breeden [1979] and Stulz [1981] which use the world market portfolio to price assets. Cosset [1984] found that the common element of variance among currencies has a very low correlation with the world market index. Therefore, models based on the world market portfolio do not satisfactorily approximate the market elements of currencies. Alternatively, the market index of the Roll and Solnik model captures the common variance among currencies well

since it is mainly composed of foreign exchange assets. Thus the market model is derived from an assumed underlying pure foreign exchange asset pricing model provided by Roll and Solnik [1977].

$$A_{j,t} = \tilde{R}_{j,t} - \hat{\alpha}_j - \hat{\beta}_j \tilde{R}_{m,t}, \quad (5)$$

where $\hat{\alpha}$ and $\hat{\beta}$ are parameter estimates derived from an OLS regression over estimation period (-55, -6) data.

$$\tilde{R}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j \sum_{i=1}^6 w_i \tilde{R}_{i,t} + \xi_{j,t}, \quad (6)$$

where $w_i = 1/6$, resulting in an equal weighting of the six foreign currencies, $i \neq j$.

Test-Statistics

The excess returns generated from the four measures are each examined to determine the significance of event period excess returns for each sample. The test-statistic is

$$T = \bar{A}_t / \hat{V}_t, \quad (7)$$

where

$$\bar{A}_t = 1/50 \sum_{i=1}^{50} A_{i,t},$$

$$\hat{V}_t = \left(\frac{\left[\sum_{t=-55}^{t=-6} (\bar{A}_t - A')^2 \right]}{50} \right)^{1/2},$$

$$A' = 1/50 \sum_{t=-55}^{t=-6} \bar{A}_t,$$

$\bar{A}_t$ is the mean daily abnormal return for day t over the fifty observations in each sample while $\hat{V}_t$ is the estimated daily standard deviation. A' is the average abnormal return over the estimation period for the entire sample. The test-statistic follows the notation used by Brown and Warner [1985] and has wide usage in other event studies of securities returns. The test-statistic is distributed Student- t under the null hypothesis of no

abnormal returns if the A_t are independent, identically distributed, and normal. In reality, the distributions of the abnormal currency returns may not be normal. In that case, other non-parametric statistical tests (such as Sign and Wilcoxon tests) need to be used. Whether the abnormal returns are normally distributed will be discussed later.

Induced Abnormal Critical Day Performance

Following the procedure of Brown and Warner [1980, 1985], abnormal performance is artificially introduced into the returns on the event day. A constant shock is added to the critical day 0 return for each of the 50 draws in each of the 100 samples used in a given test. The standard test employs a shock of plus .002 or .2%. This shock level is chosen as the standard level because we find that it generally has a high discriminatory power while not leading to a rejection of the null hypothesis in all tests at most levels of significance tests. This is consistent with Brown and Warner's [1985] selection of a 1% shock in their daily returns study. To adjust the mean return for the additional shock, 1/61 of the shock is subtracted from each of the 61 daily returns that comprise the estimation and event period.

Under the experimental design, various simulations are conducted to test the power of the four return-generation models in detecting abnormal performance under a host of experimental conditions. The first simulation module of fifty-six tests adopts a set of commonly expected experimental conditions. The results of this module of tests will be used as a basis of comparison with the results of later sets of simulations where specific experimental conditions are modified or substituted. In the standard case, the shock level induced on the event day is .2%. One hundred samples, each of which consists of fifty return observations, are used. The length of the estimation period is fifty days and missing days are allowed within the event period except on days -1 and 0. Similar to the treatment of Brown and Warner [1980, 1985], Mondays and days after holidays are treated as ordinary trading days with no return adjustment for holidays. These experimental conditions are relaxed or modified in section five to see whether the changed conditions will have a significant impact on the results. Before reporting the simulation results, we first examine the properties of the excess returns on event dates when no abnormal performance is introduced.

PROPERTIES OF EXCESS RETURNS

Table 1 shows the properties of the raw and excess returns on the event dates when the U.S. dollar is the numeraire and the deutsche mark is the foreign currency. The table is divided into two major portions: individual returns and sample-wide mean excess returns. When the raw returns and the excess returns are taken individually, there are 1132 observations and the distributions are highly non-normal.⁶ The studentized ranges all exceed 7.80, the value for the .99 fractile of the studentized range of samples

TABLE 1
Properties of Excess Returns (US\$ as Numeraire
and DM as Foreign Currency)

Models	Mean	Standard Deviation	Skewness	Kurtosis	Student Range
Individual Returns					
Raw Returns	.00004	.00710	.126	9.206**	14.00**
Excess Returns					
Random Walk	-.00005	.00726	-.567**	11.837**	13.69**
Mean Adj.	-.00009	.00736	-.692**	12.394**	13.72**
Mkt. Adj.	-.00003	.00225	-.181*	12.897**	14.77**
Market	.00002	.00189	.323**	8.871**	11.95**
Sample-Wide Mean Excess Returns					
Each Sample Consists of 30 Events					
Random Walk	.00011	.00130	.189	3.600	5.59
Mean Adj.	.00003	.00131	.052	3.649	5.46
Mkt. Adj.	.00012	.00052	.228	3.295	5.65
Market	.00015	.00044	.149	2.390	4.66
Each Sample Consists of 50 Events					
Random Walk	-.00015	.00093	-.124	3.055	4.87
Mean Adj.	-.00020	.00099	-.169	2.962	4.68
Mkt. Adj.	-.00003	.00038	-.018	3.223	5.20
Market	.00000	.00034	-.134	3.009	5.14

*Significantly different from the expected value for a normal distribution at .05 level.

**Significantly different from the expected value for a normal distribution at .01 level.

The significance test for skewness is two-tailed.

The returns and excess returns are measured at the event date '0'. No abnormal performance is artificially introduced.

There are 1132 observations for the individual returns. To examine the properties of sample-wide mean excess returns, 100 samples each consisting of 30 or 50 randomly selected events are used.

drawn from a normal population. Most of the values of skewness and kurtosis also exceed the value of the .99 fractile of the respective distribution under normality.

On the contrary, when individual returns are collected within a sample and the test is on the sample-wide mean excess return, the distribution derived from 100 randomly selected samples is close to normality. None of the skewness, kurtosis and studentized range statistics are significantly different from those of a normal distribution at the .05 level. This is true when the sample size is thirty. As the sample size is increased to fifty, which is the size used in the standard simulation, the distribution is even closer to normality. Furthermore, the means and standard deviations of the excess returns of the market-adjusted and market models decrease as the sample size grows. Smaller values indicate higher accuracy in predicting the returns. Of the four models, the market model is the best since its excess returns have the lowest mean and standard deviation.

We also examined the properties of excess returns when the foreign currency and the numeraire were varied across different currencies with the

sample size kept at fifty. The table supporting this information, together with additional supporting tables for results reported elsewhere in the article, are available from the authors upon request. In the majority of the cases, the distributions are close to normal. At the .01 level, out of 124 possible cases (4 abnormal returns models $\times$ 8 numeraires $\times$ 7 currencies) there were 14 skewness, 6 kurtosis, and 3 studentized range values significantly different from a normal distribution. Most violations occur among currencies of the European Monetary System (hereafter referred as EMS currencies), while isolated cases of violation are found among currencies of the U.S., Canada and Japan. This may be due to the fact that, from time to time, European currencies (particularly the French franc) are subjected to central bank intervention. Given the finding that some of the distributions are non-normal, both the t -test and the Wilcoxon signed-rank test are applied to some of the simulations conducted in the fifth section. The results indicate that the non-parametric signed-rank test performs slightly better than the parametric t -test with regard to both Type I and II errors. However, most often the differences are minor and the same conclusions are drawn. Although the assumption of normality may be violated, the t -test is robust enough to detect the absence or presence of abnormal performance. For the sake of brevity, unless stated otherwise, the rejection rates reported in the simulation results throughout the paper are those of the t -test.

SIMULATIONS AND RESULTS⁷

Simulations 1: Varying Currencies

Simulation 1.1: Deutsche Mark as Foreign Currency and Dollar as Numeraire

A simple simulation using the dollar as the numeraire and the deutsche mark as the foreign currency is first conducted. This simulation compares the power of the four return-generating methods in detecting abnormal currency returns. Table 2 shows the frequency with which each of the four models rejects the null hypothesis of no abnormal performance. Percentages of rejection are reported for each day of the event period (-5 to +5) at four levels of statistical significance (.05, .01, .005 and .001).

Of the four models, the market model has the best performance. Under the standard .2% shock level, the rejection rate of the market model is 100% at the .01 level. The market-adjusted model ranks second with a satisfactory score of 95% rejection. On the contrary, the random walk and the mean-adjusted models poorly discriminate the existence of abnormal returns. Only 8% and 10% of the 100 samples at the .01 level reject the null. The poor performance of the mean-adjusted model is contrary to the result of Brown and Warner [1980, 1985] where the performance of the naive mean-adjusted model is similar to that of the market model in many cases.⁸

The market model performs the best with respect to limiting Type II error (accepting a null hypothesis when it is false). By examining the rejection

TABLE 2
Deutsche Mark as Foreign Currency and Dollar as Numeraire
(Percentages of the Samples with Rejected Null Hypothesis*)

Model	Significance Level	Days within the Event Period										
		-5	-4	-3	-2	-1	0	+1	+2	+3	+4	+5
Random Walk	.05	0%	0	0	2	5	31	2	0	0	1	1
	.01	0	0	0	0	1	8	0	0	0	0	0
	.005	0	0	0	0	1	6	0	0	0	0	0
	.001	0	0	0	0	0	0	0	0	0	0	0
Mean Adj.	.05	1	0	0	3	5	34	2	0	0	1	1
	.01	0	0	0	0	1	10	0	0	0	0	0
	.005	0	0	0	0	1	8	0	0	0	0	0
	.001	0	0	0	0	0	0	0	0	0	0	0
Mkt. Adj.	.05	0	1	0	5	2	100	1	0	0	0	1
	.01	0	0	0	0	1	95	0	0	0	0	0
	.005	0	0	0	0	1	93	0	0	0	0	0
	.001	0	0	0	0	0	68	0	0	0	0	0
Market	.05	0	2	0	6	2	100	2	0	0	2	1
	.01	0	0	0	2	0	100	0	0	0	1	0
	.005	0	0	0	1	0	99	0	0	0	1	0
	.001	0	0	0	0	0	96	0	0	0	0	0

*A one-tailed test is used. The level of induced abnormal performance (shock) is .2% on the event day (i.e., day 0).

rates on the non-critical days (-5 to -1 or +1 to +5) in Table 2, the Type I errors of the models can be examined. On these non-critical days, the levels of rejection are low regardless of which model is used. The percentages are generally within the range of rejection rates that should be expected by chance at each of the significance levels. Since the market model has small Type I and II errors, it is the most dependable event study error-generating model in the dollar/deutsche mark market.

Simulation 1.2: Varying the Foreign Currency

The set of simulations of this section retain the earlier set of experimental conditions except that each of the three additional currencies is examined in separate simulations of 100 samples by varying the foreign currency. The deutsche mark results are very similar to the results of the remaining EMS currencies. For brevity, results on only the deutsche mark are reported. The dollar is kept as the numeraire throughout. The dollar is commonly used as the numeraire in research since it is the most important key currency in world trading.

The relative rankings among the four models in Table 3 remain unchanged; the market model is consistently the best among the four. The market-adjusted model generally outperforms the random walk and the mean-adjusted models except in the case of the Canadian dollar. The market-adjusted model implicitly assumes that the systematic risk of the currency is equal to one. However, the beta of the Canadian dollar is very small, having a value of .09 in the standard simulation. Using the market return as the

TABLE 3
Varying the Foreign Currency with US as Numeraire*

Significance Level	Foreign Currency											
	CD				DM				JY			
	.05	.01	.005	.001	.05	.01	.005	.001	.05	.01	.005	.001
Excess Return Models												
Random Walk	99%	96	95	72	31	8	6	0	44	11	7	1
Mean Adj.	99	96	93	65	34	10	8	0	37	11	7	1
Mkt. Adj.	47	16	9	1	100	95	93	68	64	26	19	6
Market	99	98	96	79	100	100	99	96	67	35	19	6
									85	53	42	14

*The figures in the cells are percentages of the 100 samples where the null hypothesis is rejected. .05, .01, .005 and .001 at the top of the table refers to the levels of statistical significance at which the null hypothesis is rejected. One-tailed test is used. The level of induced abnormal performance (shock) is .2%.

expected return of the Canadian dollar is therefore inappropriate and results in lower rejection rates. However, when the expected return is adjusted for the currency beta as in the case of the market model, a high rejection rate is restored for the Canadian dollar.

The free-floating currencies such as the Japanese yen and British pound have far lower levels of rejection (35% for JY and 53% for UK at the .01 level for the market model) than the EMS currencies (ranging from 76% for BF to 100% for DM and NG). The Canadian dollar, normally regarded as free-floating, has high rejection rates since its value is closely linked to the dollar. The observations above can be partially explained by moderate daily fluctuations of the EMS currencies resulting from close governmental intervention while free-floating currencies have more volatile daily fluctuations. Simulation 1.5 considers the impact that the beta and selection of market index also have on providing the above results. Though rejection rates on non-critical days are not shown, they are found to be minimal. This finding is consistent throughout most simulations of this study.

The findings above indicate that the market model is usually the preferred model to be used. The random walk or the mean-adjusted model may be the most appropriate when the foreign currency and the numeraire are closely linked. Event studies using a free-floating currency as the foreign currency may require a higher shock level in order that abnormal currency returns may be effectively detected. Whether this is true or not will be examined under Simulation 2.1.

Simulation 1.3: Varying the Numeraire

Different foreign currencies are now allowed to be the numeraire currency in this simulation. The findings will be useful in examining event studies between two currencies, neither of which is the dollar. Table 4 presents the percentages of rejection of the null hypothesis using different currencies as numeraires and foreign currencies. The .01 significance level is used to report rejection since it has a high discriminatory power by recognizing different rejection levels over different numeraires and foreign currencies. Given seven foreign currencies and five numeraires, taking two at a time, there are a total of thirty-five simulations shown in Table 4.

Though different currencies are used as the numeraire, the relative rankings among the four models are more or less unchanged. The market model consistently outperforms the other three models. The random and the mean-adjusted models have unsatisfactory rejection scores that are lower than the market-adjusted model except when the numeraire currency and the foreign currency are closely linked (e.g., US/CD and the unreported EMS currencies; hereafter, the abbreviation X/Y is used to represent an event study using X as the numeraire and Y as the foreign currency).⁹

When the foreign currency is one of the EMS currencies, represented in Table 4 by the deutsche mark, the rejection rates are usually high. The impact of the market weighting scheme in contributing to this phenomenon is covered in section five, Simulation 5. Among the free-floating currencies,

TABLE 4
Rejection Rates on the Critical Day over Combinations
of Selected Numeraires and All Foreign Currencies*

Numeraire	Foreign Currency							
	US	BF	CD	DM	FF	JY	NG	UK
US	Random Walk	11%	96	8	9	11	11	13
	Mean Adj.	12	96	10	12	11	11	18
	Mkt. Adj.	65	16	95	79	26	96	50
	Market	76	98	100	88	35	100	53
CD	Random Walk	3		13	7	8	10	18
	Mean Adj.	3		13	8	9	11	18
	Mkt. Adj.	53		94	76	23	94	56
	Market	67		100	88	28	98	58
DM	Random Walk	13	9		80	24	100	41
	Mean Adj.	15	8		83	24	100	41
	Mkt. Adj.	29	34		58	33	81	54
	Market	70	64		85	45	100	62
JY	Random Walk	4	8	19	9		26	20
	Mean Adj.	8	13	24	11		31	16
	Mkt. Adj.	19	22	87	73		87	37
	Market	29	19	97	76		95	40
UK	Random Walk	9	16	31	27	17	39	
	Mean Adj.	9	15	27	25	14	36	
	Mkt. Adj.	23	28	84	67	25	81	
	Market	34	28	92	75	22	86	

*The figures in the cells are percentages of the 100 samples where the null hypothesis is rejected at the .01 significance level. One-tailed test is used. Induced abnormal performance (shock) is .2%.

the yen has the lowest rejection rates, which are around 20% or 30% for the market model. This is probably due to the higher volatility of the yen. The relationship between the foreign currency and the numeraire is not necessarily reciprocal. Although the yen does not perform well as a foreign currency, it is not a poor numeraire currency. For instance, when the deutsche mark is the numeraire and the yen is the foreign currency, the rejection rate is 45% for the market model at the .01 level. However, when the yen is the numeraire and the deutsche mark is the foreign currency, the rejection rate is 97%.

Relative to the other currencies, the U.S. dollar is found to be fairly dependable numeraire currency, with an average rejection rate of 79% for the market model. This finding should be of interest to researchers who have support for the continued use of the dollar as the numeraire in event studies.

Simulation 1.4: Samples with Multiple Foreign Currencies

Event studies may often involve more than two currencies. For instance, the study of Cosset and Doutriaux de la Rianderie [1985] examines whether foreign exchange markets responded efficiently to the announcement of political events. Their sample includes events that span seven countries. To meet the needs of this kind of research, a simulation is performed by randomly selecting the draws with equal probability from the seven foreign currencies, with respect to a selected numeraire.

The overall rejection rates lie somewhat between the high and the low rates reported previously in Simulation 1.3. The relative rankings of the four return-generating models are once again confirmed. The market model is consistently the best of the four. As far as the choice of numeraire currency is concerned, the U.S. dollar has the highest rejection rates, scoring 83% for the market model at the .01 level, and is therefore a reasonable numeraire to select for event study tests.

Simulation 1.5: Market Index Affected by Shocks

In the foreign exchange market, the market is made up of a few frequently traded currencies. When a selected foreign currency is shocked, it already represents a significant proportion of the market portfolio. Furthermore, there is currency risk clustering when more than one other currency in the market basket responds to the event. The market model may lose its power in detecting abnormal changes in this environment. Since a likely objective is to determine unexpected returns associated with an event, a procedure that nets out any systematic return changes across several currencies may be inappropriate. Since the market-adjusted and market models are intentionally designed to net out systematic effects, they may be only appropriate for use in the foreign currency markets when the empiricist is trying to restrict abnormal returns detection to only non-systematic changes between the numeraire and specific foreign currency. The random walk and mean-adjusted excess return models are more appropriate if the objective is to measure the entire unexpected return.

The differences in results between the systematic risk-based models and non-systematic risk-based models are strongest when examining foreign currencies heavily weighted in the market index. The heavy weighting may come from risk clustering, as with the EMS currencies, or from dominant countries when a size-weighted index is used (for example the U.S. dollar as a foreign currency with a GNP-weighted market index). Differences in results between the systematic and non-systematic models decrease as the foreign currencies correlation with the index, and thereby its beta, approach zero. Thus the market model collapses to the mean-adjusted model when the beta equals zero. The superiority of the market model noted in prior simulations provides evidence that better detection of idiosyncratic shocks occurs. Evidence that this no longer holds when the currency has a high beta and the shock is systematic is now examined.

An example of the above problem concerns the European Monetary System (EMS) currencies. The member countries of the EMS are located in the same region and there is an exchange rate agreement among the members. When one of the EMS currencies is selected to be the foreign currency, it is possible that the other three EMS currencies, which are also included in the market basket, will also be affected by the event at various degrees. The market index will then be substantially affected by the abnormal performance, as will the EMS currency. The EMS currency will have a high beta since the index is composed of an equal weighting of six currencies, of which three or four are EMS currencies.¹⁰

To simulate the consequences of systematic shocks, the selected foreign currency is still shocked at .2% level on the event date while the market index is given a shock level randomly selected from a uniform distribution from 0% to .4% with an expected value of .2%.¹¹ The random shock is used since the researcher may not be sure to what extent the market index is affected. Furthermore, to examine how the performance of the models may deteriorate when the market index is increasingly affected by the shock, a weight factor of 0.2, 0.5, 0.8 or 1.0 is multiplied to the shock. When the weight factor is 1.0, all the currencies in the market basket are affected. When the weight is 0.5, it may represent the case when half of the currencies are affected at the same expected shock level of .2%. An example is the situation when only the set of EMS currencies reacts simultaneously to the same event, since EMS currencies constitute about half of the market basket in this study.

The performance of the market and market-adjusted models deteriorates as the weight factor increases. At the .2 weight level, the market model still has the best performance among the four. When the weight is increased to .5, the performance of the market and market-adjusted models becomes inferior to that of the random and mean-adjusted models in most cases. Consistent with the explanation above, deterioration is not found when the numeraire and the foreign currency are closely tied (e.g., US/CD, $\beta = .09$; DM/NG, $\beta = .127$). In such cases, the market model maintains a satisfactory

level of performance even when the weight factor is increased to 1.0.¹² When the numeraire and the foreign currency are closely tied the exchange rate between the two currencies is heavily influenced by factors other than the marketwide movements. The beta of the foreign currency with respect to the market index will be small on these closely tied currencies. Thus, the value of the expected return for the foreign currency will not depend much on the market index. As a result, even when the market index is shocked to the full .2% level, a satisfactory level of performance can still be maintained for this particular foreign currency. As expected, the results show that the performance of the random walk and the mean-adjusted models remain unchanged since their excess return measures are independent of the market index.

Under circumstances where it is difficult to know in advance whether an event will have systematic (clustering) effects on several other currencies, a two-stage approach would be appropriate. First, the mean-adjusted or random walk model may be used to detect whether significant abnormal performance is found in other currencies that constitute the market basket (using the same numeraire). If substantial abnormal performance is found in several other currencies, the market index will be seriously affected by the shock and currency risk clustering will reduce the power of market-adjusted and market models. The mean-adjusted or random walk model for the foreign currency should be adopted. However, if no or little abnormal performance is found in other currencies, the researcher may use the market model to test the selected foreign currency since it is the most powerful model in detecting non-systematic abnormal performance.

Simulations 2: Varying Shock Levels, Time Periods and Variance

Simulation 2.1: Varying Shock Levels

The following simulations vary the shock levels from 0%, .2% to .5% to study the effects and report results at the .01 significance level. The 0% shock is used to confirm the level of Type I errors of the four models. At the .5% shock level, even the more volatile Japanese yen has rejection rates close to 100%. Additional simulations show that a shock level of .3% may be sufficient for most free-floating currencies to attain satisfactory scores. These findings indicate the levels of abnormal performance required in real data in order to be effectively detected. In the case of EMS currencies, except the Belgian franc, the .2% level is sufficient. Floating currencies such as US, CD, JY and UK may need a shock level of from .3% to .5%. When the shock level is zero, the rejection rates of the four models follow expectations and are mostly zero or 1%, confirming that Type I error is not substantial regardless of which model is used.

A supplemental simulation examined the impact of varying shock levels by splitting the time-series data into two equal subperiods. The test was motivated by the increasing volatility in foreign exchange markets occurring since the advent of the floating rate system in 1973. For each given shock

level, the rejection rates are close to each other between the two time periods. The rejection rates in the second period are slightly lower for the non-European free-floating currencies (US, CD and JY) and slightly higher for the European currencies (EMS currencies or UK).

Simulation 2.2: Increase of Variance in the Event Period

Studies in the stock market show that substantial variance increases occur around many types of events (see, for example, Beaver [1968] and Patell & Wolfson [1979]). This potential phenomenon in the foreign exchange market is examined by doubling return variance. This follows the Brown and Warner [1985] approach. The bias in rejecting the null hypothesis of no abnormal returns when variance has increased is then determined. Each critical day 0 return is transformed

$$\tilde{R}_{j,t} = \tilde{R}_{j,t} + (\tilde{R}_{j,t-5} - \bar{R}_j), \quad (8)$$

where the $\tilde{R}_{j,t-5}$ day return is arbitrarily selected. $\bar{R}_j$ is defined in equation (3) as the estimation period average return for currency j and $\tilde{R}_{j,t}$ is defined in equation (1). This approach results in effectively doubling the day zero variance.

As expected, the results show that the variance increase reduces the sensitivity of the event study methodologies. When a .2% shock is induced on the event day, the rejection rates are generally smaller, thereby signalling a higher Type II error. For instance, when the numeraire is the Japanese yen and the foreign currency is the deutsche mark, the rejection rate decreases from 97% to 73% for the market model at the .01 level. When there is no shock on the event day, the Type I error of rejecting the null hypothesis is higher. Such an increase is more obviously seen when the foreign currency is an EMS currency and the numeraire is a free-floating currency. For instance, when the numeraire is the U.S. dollar and the foreign currency is the deutsche mark, the rejection rate increases from 0% to 22% for the market model at the .01 level. The above findings provide a warning that if a variance increase occurs around the event date, Type I and II errors are likely to increase. Researchers would be encouraged to test for this phenomenon and adopt appropriate statistical tests to accommodate any event period variance increase.

Simulations 3: Sample Size, Length of Estimation Period, Missing Days on Non-Critical Days

Simulation 3.1: Varying Sample Size

In the standard test, each sample consists of 50 draws. However, in actual event studies, researchers may have difficulty in finding 50 events. The following simulations examine sample sizes of 30, 50 and 100 to determine the impact of sample size on the sensitivity of the methodologies. There is indeed improvement when the sample size is increased from 50 to 100

(e.g., US/BF: 76%→98% for the market model at the .01 level). The improvement is probably due to both an apparent decrease in sample variance noted in the Properties of Excess Returns Section and the lower *t*-values required for the same .01 level of statistical significance. Alternatively, when the sample size is reduced to 30, the drop of rejection rates is less obvious in some cases (e.g., US/DM: 100%→99%) and substantial in other cases (e.g., UK/FF: 88%→62%). Of the eight foreign currencies, the effect of sample size is most obviously seen when the Japanese yen is the foreign currency; the rejection rates generally drop to less than 20%.

Identification of more relevant events, in order to enlarge the sample size, materially improves the likelihood of obtaining reliable results. While a sample size of thirty may still be acceptable for the deutsche mark or Netherlands guilder, it may be too small when floating currencies are used as foreign currencies. If a sufficiently larger sample cannot be obtained, the level of statistical significance may have to be reduced to 5% or 10%.

Simulation 3.2: Allowing Missing Days on Non-Critical Days

The restriction of no-missing-days within the eleven-day event period (−5, +5) may substantially reduce the sample size which in turn reduces the sensitivity of the methodology. However an event period of eleven days or more is not uncommon when examining for possible abnormal returns in the vicinity of the critical day, or when the precise critical day is not known. Removing the restriction seems to result in higher rejection rates when floating currencies serve as the foreign currency (e.g., US/JY: 37% vs. 7% for the market model at the .01 level). When the foreign currency is an EMS currency, the relationship is less obvious: some are better off (e.g., US/DM: 100% vs. 97%), while others are worse off (e.g., US/BF: 76% vs. 85%). The Belgian franc is the only foreign currency that shows a clear disadvantage when the restriction is relaxed. Such findings signal that it is generally acceptable to relax the restriction of no-missing-days within the event period to gain better estimation when there are more events in the sample.

Simulation 3.3: Varying Length of Estimation Period

Generally, longer estimation periods provide better estimates due to an amelioration of idiosyncratic errors. This may not hold if frequent parameter shifts occur over time. The length of the estimation period is varied from 30, 40, 50 to 60 days in this set of simulations. The minimum number of days that must have full information for retention is 20, 25, 30 and 30 days respectively.

Table 5 reports the results for each of four models. Although differences are seen among the four columns of rejection rates, the differences are not large. This finding indicates that when there are practical constraints, a shorter estimation period of thirty days is often acceptable. Longer estimation periods will also not necessarily lead to higher rejection rates. The optimal length seems to vary across currencies. For instance, when the

French franc is the foreign currency, a shorter estimation period of thirty days may be optimal; in the case of the deutsche mark, a length of forty or fifty days generally leads to higher rejection rates. Table 5 might be used as a guide to determine the optimal length when actual event studies are conducted.¹³

Simulation 4: Interest-Rate Adjustments

Mondays and trading days after holidays are generally treated as ordinary trading days without return adjustment in the stock market event studies. The standard test of this study follows this general practice. This treatment may be inappropriate in foreign exchange markets. When foreign currencies are bought, they can be deposited in banks to earn interest and interest is paid over holidays and weekends. The question is whether there is a significant difference between the easier, conventional approach and the more appropriate approach where weekend and holiday returns are adjusted. The simulations in this section examine the effects of four different adjustments of the interest rates. In comparing the results of the different adjustments, two questions are investigated: 1) whether the inclusion of the interest-rate differential is crucial in calculating the currency return, and 2) whether the estimation of returns for non-trading days improves the power of the methodology.

Procedures

In the first and the second adjustments, Mondays and days after holidays are treated as ordinary trading days. In the third and fourth adjustments, returns are estimated for non-trading days.

In the first procedure, adopted in all prior simulations, the interest rates are included in the currency return calculation and no return adjustment is made for days after holidays. The second adjustment leaves out the interest rates from the equation when currency returns are calculated. If this treatment works well, effort required in identifying Mondays and days after holidays for special adjustment can be saved. More importantly, foreign exchange event studies previously constrained by data problems would be permissible. Foreign exchange market data on daily interest rates are not easily available. That is why the present study is confined to only eight major currencies. Currencies of other countries, especially those of less developed countries, may not have obtainable daily interest rates; or, if the data exist, they are difficult to collect. If interest rates are not crucial in determining the existence of abnormal returns, a substantial burden currently facing researchers who wish to examine foreign exchange markets would be removed.

In the third adjustment, returns are estimated for non-trading days. When the numeraire or the foreign currency is not traded on a particular day, or daily interest rate is not available for either of the currencies, a geometric mean daily return is calculated with the following equation:

TABLE 5a
Varying the Length of Estimation Period

Numeraire	Foreign Currency																							
	US						BF						CD						DM					
	30	40	50	60			30	40	50	60			30	40	50	60			30	40	50	60		
US							12%	14	11	6			94	98	96	97	11	9						
							9	12	12	7			95	96	96	96	7	11	10	11				
							38	56	65	65			24	18	16	16	12	76	97	95	98			
							56	76	76	78			98	99	98	99	98	100	100	100	100			
BF	10	13	11	7									20	20	9	13	95	81	77	71				
	10	15	12	5									23	20	7	12	95	75	70	74				
	11	28	25	15									18	37	21	27	73	77	69	62				
	47	52	55	29									38	50	38	32	97	100	99	97				
CD	85	98	96	99			11	10	3	9							12	13	13	11				
	93	99	97	100			9	11	3	10							8	10	13	11				
	20	17	16	16			38	57	53	59							86	97	94	96				
	96	100	98	100			43	70	67	75							99	100	100	100				
DM	12	13	13	6			35	64	58	68			24	13	9	10								
	19	17	15	5			39	72	61	74			55	12	8	9								
	19	31	29	18			29	45	40	40			30	38	34	26								
	87	76	70	59			43	76	72	74			84	67	64	53								
FF	2	14	8	16			50	54	52	61			11	18	9	14	99	91	99	92				
	5	13	15	6			51	64	61	68			12	20	8	9	99	90	98	92				
	11	27	21	29			24	24	34	33			13	32	26	31	73	29	72	70				
	37	46	35	50			60	65	69	75			31	55	33	34	100	99	99	99				
JY	6	7	4	4			10	11	20	11			12	10	8	7	3	23	19	13				
	31	12	8	5			25	16	25	13			30	13	13	10	17	31	24	21				
	20	28	19	10			51	68	55	54			45	22	22	19	68	86	89	85				
	40	38	29	20			80	61	67	66			34	49	19	23	83	92	97	91				
NG	23	13	14	9			31	58	65	69			19	10	15	20	100	100	100	100				
	26	16	15	9			31	65	74	75			22	12	13	15	100	100	100	100				
	32	28	37	31			18	47	49	41			28	33	35	39	63	67	79	68				
	78	64	68	62			27	73	80	79			86	59	55	54	100	100	100	100				
UK	6	12	9	9			1	18	19	16			4	20	16	7	5	29	31	21				
	13	16	9	9			7	23	21	18			9	22	15	4	9	29	27	21				
	29	32	23	24			27	46	43	37			43	39	28	23	78	82	84	77				
	42	32	34	25			29	64	61	54			36	39	28	19	78	95	92	86				

TABLE 5b
Varying the Length of Estimation Period*

Numeraire	Foreign Currency																	
	FF						JY						NG					
	30	40	50	60	70	80	30	40	50	60	70	80	30	40	50	60	70	80
US	13%	10	9	9	11	11	27	12	11	15	13	11	35	16	13	13	17	17
	11	11	12	11	11	11	15	7	11	10	11	10	21	14	13	14	17	17
	82	76	79	76	22	27	22	27	26	25	80	96	77	48	50	48	50	41
	89	88	88	89	21	36	21	36	35	30	98	100	74	60	53	54	54	54
BF	88	66	60	59	19	16	19	16	21	12	93	76	71	34	36	22	22	22
	87	66	56	64	6	15	6	15	16	9	93	68	47	30	33	23	23	23
	51	55	52	44	11	23	23	23	23	18	65	73	70	46	45	25	25	25
	95	84	87	86	5	20	29	29	29	14	95	97	59	66	59	40	40	40
CD	13	10	7	7	23	12	23	12	8	7	18	12	34	17	18	7	7	7
	16	10	8	11	24	10	24	10	9	8	13	14	15	20	18	10	10	10
	79	76	76	71	55	31	55	31	23	21	88	93	85	46	56	24	24	24
	90	81	88	86	74	36	74	36	16	28	97	97	72	55	58	33	33	33
DM	95	83	80	91	52	19	52	19	24	29	100	100	73	47	41	25	25	25
	95	84	83	92	41	21	41	21	24	23	100	100	54	49	41	26	26	26
	74	49	58	54	72	28	72	28	33	32	59	68	81	53	54	34	34	34
	96	88	85	90	74	38	74	38	45	32	99	100	81	71	62	53	53	53
FF					27	28	27	28	18	18	88	98	60	32	37	25	25	25
					20	25	20	25	15	15	97	88	32	31	35	26	26	26
					55	39	55	39	26	26	78	71	42	85	51	35	35	35
					52	45	52	45	31	34	100	100	75	63	63	50	50	50
JY	13	12	9	13							7	25	18	12	10	9		
	29	16	18	11							21	30	35	18	16	12		
	94	79	73	73							75	91	85	52	37	42		
	93	78	76	81							85	88	84	65	40	46		
NG	89	84	77	74	41	23	41	23	20	18			75	39	27	27		
	88	86	82	77	33	22	33	22	16	14			46	46	34	29		
	65	59	65	46	65	33	65	33	31	28			82	40	49	26		
	98	90	88	84	59	38	59	38	30	36			78	60	54	57		
UK	1	19	27	26	16	11	16	11	17	18	3	29						
	6	20	25	22	19	9	19	9	14	15	8	23						
	64	51	67	65	84	27	84	27	25	39	72	81						
	69	61	75	75	79	26	79	26	22	34	84	87						

*The figures in the cells are percentages of the 100 samples where the null hypothesis is rejected at the .01 significance level. The four rows of percentages are those of the random walk model, the mean-adjusted model, the market-adjusted model and the market model, respectively. The level of induced abnormal performance (shock) is .2%. One-tailed test is used. 30, 40, 50 and 60 at the top of the table represent different lengths of the estimation period in number of days.

$$\bar{R}_{j,i} = \left(1 + \frac{S_{j,v} - S_{j,t}}{S_{j,t}} - (v-t)(r_{n,t} - r_{j,t}) \right)^{1/(v-t)} - 1, \quad (9)$$

for $i = t+1, t+2, \dots, v-1$ where missing information on exchange rates or rates of return occur from $t+1$ through $v-1$.

The fourth adjustment method also estimates returns for non-trading days. However, the interest rates are left out from the equation when currency returns are calculated. The equation is similar to equation (9) except for the interest-rate component:

$$\bar{R}_{j,i} = \left(1 + \frac{S_{j,v} - S_{j,t}}{S_{j,t}} \right)^{1/(v-t)} - 1. \quad (10)$$

The averaging procedure of the third and fourth approaches may lead to a biased underestimate in variance and therefore a biased overstatement in calculated t -statistics. It is, however, worth a trial examination to see if the adjustment provides unbiased results. If this adjustment works well, it will substantially reduce the problem of sample size. In the standard test, quite a few event dates need to be excluded because one of the eight currencies does not have exchange rate or interest-rate information. Even though that currency may not be the numeraire or the foreign currency, its information is still needed in the calculation of market returns for the market-adjusted and market models.

Results

Table 6 reports four columns of rejection rates in each cell, representing the four adjustments respectively. Comparing the figures of the first two columns, elimination of interest rate in the calculation of returns actually yields improved results (e.g., DM/US: 70%(I)→86%(II); JY/CD: 19%(I)→54%(II) for the market model). This indicates that when the interest-rate information is incorporated, additional idiosyncratic variance may be induced and, therefore, a reduction in t -statistics results when abnormal returns are present.

However, when the interest-rate information is incorporated and adjusted to represent missing days, the power of the methodology is improved. This can be seen by comparing the rejection rates of the third and fourth columns. When returns are estimated for non-trading days, the third adjustment method (interest rates included in return calculation) outperforms the fourth method (interest rates excluded). Nevertheless, the differences are small. Because of the small differences, it is generally acceptable to neglect interest rates in the calculation of currency returns. Furthermore, the rejection rates of the last two columns are generally higher than the first two. This result indicates that estimating returns for non-trading days will also enhance the power of the methodology. Checking the rejection rates on non-trading days will also enhance the power of the methodology.

TABLE 6
Interest-Rate Adjustments*

Numeraire	Foreign Currency															
	US				CD				DM				JY			
	I	II	III	IV	I	II	III	IV	I	II	III	IV	I	II	III	IV
US					96	100	100	100	8	25	42	46	11	35	51	52
					96	100	100	100	10	22	43	41	11	23	48	47
					16	26	52	45	95	100	100	100	26	62	74	79
					98	100	100	100	100	100	100	100	35	57	74	78
CD	96	100	100	100					13	32	51	54	8	27	42	45
	97	100	100	100					13	27	54	50	9	23	39	31
	16	21	73	59					94	99	100	100	23	53	76	80
	98	100	100	100					100	100	100	100	28	46	99	78
DM	13	18	43	38	9	21	43	32					24	39	62	56
	15	24	47	43	8	28	44	39					24	30	58	51
	29	46	74	72	34	48	77	61					33	56	78	77
	70	86	97	96	64	82	90	90					45	47	83	78
JY	4	10	44	31	8	18	46	42	19	34	70	67				
	8	19	47	44	13	29	48	44	24	42	75	69				
	19	42	68	67	22	41	77	68	89	96	98	99				
	29	48	82	79	19	54	77	74	97	96	99	100				
UK	9	18	49	49	16	31	53	48	31	61	72	80	17	36	58	59
	97	20	50	45	15	31	52	45	27	54	74	70	14	26	55	52
	23	37	68	61	28	37	74	65	84	95	98	98	25	50	72	77
	34	45	74	69	28	50	77	74	92	97	97	98	22	40	77	72

*The figures in the cells are percentages of the 100 samples where the null hypothesis is rejected at the .01 significance level. The four rows of percentages are those of the random walk model, the mean-adjusted model, the market-adjusted model and the market model, respectively. One-tailed test is used. The level of induced abnormal performance (shock) is .2%.

I, II, III and IV at the top of the table represent different interest-rate adjustment methods:

I. Mondays and days after holidays as ordinary trading days;
II. Interest rates excluded from currency return calculation;
III. Returns estimated for holidays, interest rates included;
IV. Returns estimated for holidays, interest rates excluded.

Checking the rejection rates on non-critical days of these four different adjustments, the percentages are again minimal, showing that the adjustments will not substantially increase Type I errors. Earlier, concern was indicated that the averaging procedure used by the third and fourth methods may lead to biased overstatement in calculated t -statistics. The empirical findings, however, show that the effect is not serious in increasing Type I error, and that the procedure indeed materially decreases Type II errors.

Simulation 5: Choice of Market Return Proxy

Brown and Warner [1980] examine the effects from the choice of market return and find that power in correctly predicting abnormal returns with the equally weighted index is as strong or stronger than the value-weighted index of security returns. The same phenomenon is now examined and found in the foreign exchange market. Thus far, market currency returns were derived by using an equal weighting of the six foreign currency returns, where the foreign currency being examined is omitted from the market weight calculation. Two alternative measures of deriving market return proxies are examined.

With the first procedure, GNP information from 1978 through 1987 is used to derive weights for the different currencies in calculating a weighted average daily market return. The market weighting for a given currency on any day is derived by dividing the currency's quarterly GNP figure by the sum of the quarterly GNP figures on the six foreign currencies. The set of six weights (excluding the numeraire and the foreign currency) and six daily $\bar{R}_{j,t}$ are then used to derive a weighted average daily market return. This weighting scheme is recalculated for each of the eight possible numeraire currencies.¹⁴

The Special Drawing Right, an international reserve currency created by the International Monetary Fund, is used to form the third weighting scheme. Due to the unavailability of interest-rate data on the sixteen currencies used in deriving the SDR prior to 1981, the 1981-1985 SDR weighting factors are used for the 1978-1985 time-frame of this study, while the 1986-90 weighting factors are used for this study's 1986 through February 1988 time-frame. For a given foreign currency, j , and using the U.S. dollar as the numeraire, n , in this example, the weight used for currency j in deriving its contribution to the market portfolio return is

$$w_{j,t} = S_{j,t} U_j / \sum_{k=1}^{k=5} S_{k,t} U_k, \quad (11)$$

where U_k represents the number of local currency units per SDR, the $S_{k,t}$ are the US cents per unit of foreign currency and for $k=1, 2, \dots, 5$ for the five currencies used in the SDR and $S_{US,t}=100$ (cents). The weights of the five currencies vary every day. This occurs since the currency weights

depend both on the fixed currency units, U_k , and on the daily exchange rates of the currencies. The SDR weighted market return on a given day t is then

$$\tilde{R}_{m,t} = \sum_{j=1}^{j=5} w_{j,t} \tilde{R}_{j,t} \quad (12)$$

The $w_{j,t}$ and $\tilde{R}_{m,t}$ are recalculated daily over each of 2,650 days for which there is adequate information.

Results

Regardless of which market index is used, the market model is still the most sensitive in detecting abnormal performance of currency returns. The market-adjusted model still outperforms the random walk and the mean-adjusted models except when the foreign currency is closely linked to the numeraire currency.

Among the three market indices, the equally weighted index generally outperforms the GNP- and SDR-weighted indices. The only clear exception is found when the Canadian dollar is the foreign currency where the GNP-weighted index is predominant. The rejection rates under the three alternative indices are usually close to one another. In those cases when the performance of the indices do not differ substantially, the SDR-weighted index may be an acceptable market proxy. Since the SDR consists of only five commonly traded currencies, the computation of the market return can be simpler and the data are easier to collect.

Simulation 6: Standardization of Abnormal Returns

Many of the recent event studies in stock market employ a specific procedure to standardize the abnormal returns to control for heteroskedasticity. The simulation is performed to see whether this adjustment increases or decreases the power of the event study methodology.

Procedures

The generation of daily standardized prediction errors in this section and the cumulative standardized prediction errors in Simulation 7 are computed according to Patell [1976]. First, the standardized prediction error on the event day, t , is derived. There are $N=50$ days in the variance estimation period. The standard prediction error is

$$SPE_{j,t} = A_{j,t} / V_{j,t} \quad (13)$$

where

$$V_{j,t} = \sigma_{j,t} [1 + 1/N + (\tilde{R}_{m,t} - \bar{R}_m)^2 / \sum_{i=-55}^{-6} (\tilde{R}_{m,i} - \bar{R}_m)^2]^{1/2},$$

$$\sigma_{m,t} = \{ [\sum_{i=-55}^{-6} (\tilde{R}_{j,t} - \hat{\alpha}_j - \hat{\beta}_j \tilde{R}_{m,i})^2] / (N-2) \}^{1/2},$$

$$\bar{R}_m = 1/N \sum_{i=-55}^{-6} \tilde{R}_{m,i},$$

where $V_{j,t}$ is the estimated standard error of $A_{j,t}$, $\sigma_{j,t}$ is the residual variance of draw j 's market model regression, $\bar{R}_m$ is the average market return in the estimation period for a given numeraire and a given draw. This procedure continues to be widely used. For example, see Linn and Pinegar [1988] for a recent application.

The average standard prediction error for a given sample is

$$ASPE_t = 1/P \sum_{j=1}^P SPE_{j,t}, \quad (14)$$

where $P=50$, the number of draws per sample, or observations on day t . The individual prediction errors are assumed to be normally distributed. Thus each $SPE_{j,t}$ is distributed as a student- t with variance equal to $(P-2)/(P-4)N$, and approaches $1/N$ with large P . Under the null hypothesis that the cross-sectional mean of $A_{j,t}$ is zero, the test-statistic is asymptotically unit normal.

$$Z = P^{1/2} ASPE_t. \quad (15)$$

Results

Table 7 reports the percentages of rejection of the unstandardized ('No' columns) and standardized ('Yes' columns) abnormal returns on three days of the event period: the day before the critical day, the critical day and the day after $(-1, 0, +1)$. Only the percentages of the market model are reported.

The results show that the rejection rates on the critical day are generally improved. The improvement is easily seen when the foreign currency is a free-floating currency (e.g., DM/US: 79% → 94%; US/JY: 35% → 76%). Nevertheless, the improvement of Type II error on the critical day is accompanied by an increase of Type I error. As seen from the rejection rates on non-critical days where there are no shocks. This side-effect is especially serious when the foreign currency is the deutsche mark or the Netherland guilder (not reported), where the rejection rates are already high even before the standardization procedures are applied. The findings suggest that the standardization procedure does not seem to be well suited for application in foreign currency markets.

TABLE 7
Standardization of Abnormal Returns*

Numeraire	Days	Foreign Currency									
		US		DM		FF		JY		UK	
		No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
US	-1			0	5	1	5	0	1	0	2
	0			100	100	88	100	35	76	53	94
	+1			0	1	0	3	0	5	0	1
DM	-1	0	5			2	8	0	4	1	1
	0	70	94			85	99	45	85	62	88
	+1	1	5			0	1	0	3	1	3
FF	-1	0	4	3	13			0	7	0	1
	0	50	89	99	100			34	79	63	94
	+1	0	3	1	8			0	6	0	1
JY	-1	0	0	0	3	0	8			0	2
	0	29	81	97	100	76	99			40	90
	+1	0	1	0	4	0	1			0	1
UK	-1	0	1	2	5	0	2	0	2		
	0	18	77	92	100	75	98	22	75		
	+1	0	0	1	1	0	0	0	4		

*The figures in the cells are percentages of the 100 samples where the null hypothesis is rejected at the .01 significance level.

The three rows represent the percentages of rejection of the market model on days -1, 0 and +1 of the event period.

One-tailed test is used. The induced abnormal performance (shock) is .2%.

Simulation 7: Imprecise Event Date

The simulations that have been conducted thus far all assume that the event day is known with certainty. In most research, the exact event day may not be known. One method frequently used to investigate abnormal performance under these circumstances is the Cumulative Average Residual (CAR).

Procedures

For each security, abnormal performance is randomly induced for each draw within each sample during the event interval (-5, +5). The standardized cumulative abnormal return is derived, based on the assumption that prediction errors are independent (see Patell [1976]).

$$CSPE_{T_1, T_2} = 1/P \sum_{i=1}^p \sum_{t=T_1}^{T_2} SPE_{i,t} / (T_2 - T_1 + 1)^{1/2}, \quad (16)$$

where $T_1 = -5$ and $T_2 = +5$ and equation (15) is transformed to an asymptotically unit normal variable in

$$Z = P^{1/2} CSPE_{T_1, T_2}.$$

TABLE 8
Imprecise Event Date*

Numeraire	Foreign Currency											
	US			CD			DM			FF		
	0%	.2%	.5%	1%	0%	.2%	.5%	1%	0%	.2%	.5%	1%
US	5	14	33	55	13	46	50	69	9	25	50	83
CD	0	0	2	3	1	4	11	21	15	48	82	96
DM	0	1	2	4	1	4	7	16	54	82	97	100
FF	0	0	1	3	0	0	2	6	11	21	50	74
JY	0	0	0	0	0	1	1	2	17	30	58	81
UK	0	0	0	0	0	0	0	0	3	12	32	57
									18	23	31	47

*The figures in the cells are percentages of the 100 samples where the null hypothesis is rejected at the .01 significance level on day +5.
The percentages of rejection are those of the market model given different level of induced abnormal performance: 0%, .2%, .5% and 1%.
One-tailed test is used.

Results

Table 8 reports the rejection rates of the market model on day +5, the last day of the event period. When the shock level is zero, the rejection rates are unacceptably high, indicating the existence of substantial Type I errors. The worst case is seen when the British pound is the foreign currency; the rejection rates are all two-digit percentages.

At the .2% and .5% shock levels, the rejection rates are generally low, indicating high Type II errors. When the shock level is raised to 1%, the EMS currencies (except the French franc) start to show satisfactory scores. But the free-floating currencies still do not perform well. These findings confirm Brown and Warner's [1980] findings in the stock market that knowledge of the precise event date is important.

Even when the precise event day is known, event studies in the foreign exchange market may also be affected by imprecise event time within the day since many events may be released within the day. Unlike the common stock returns which are affected by a small number of firm-specific events (e.g., earnings announcements, dividend changes, mergers and so forth), the exchange rate market is sensitive to a broad spectrum of economic, social and political news. Foreign exchange market participants may sometimes be confronted with different country-specific news on the same day. For instance, in the U.S., the unemployment rates, trade balances and the inflation rates are from time to time announced on the same Friday. Under these circumstances, it is difficult to distinguish the individual effects of these different types of news. The impact from an imprecise timing of the events within the day would lead to the same phenomenon of a deterioration in the power of the tests that we find in our multiple-day imprecise date simulation. To tackle this problem, researchers may consider only major and nonsynchronized events or try to use hourly exchange rate data.

Alternatively to finding the precise within-day release time for each time of information affecting a currency, or in case they are released simultaneously, a multivariate regression could be used to test the impact of simultaneous events on the dependent variable of abnormal returns. The value, or standardized value, of each effecting item is treated as an independent variable. Tests for multicollinearity should be made. Possible interaction variables would be incorporated into the analysis in order to enable evaluation of the impact of highly collinear variables.

SUMMARY AND CONCLUSIONS

In this study, daily currency returns were used to examine the power of event study methodologies in detecting abnormal performance in the foreign exchange market. The paper compared the performance of four alternative abnormal return models—the random walk model, the mean-adjusted returns model, the market-adjusted returns model and the market model—under different experimental conditions. Some of the issues explored

are similar to those in Brown and Warner [1980, 1985] (e.g., ranking of models, choice of market index, imprecise event date and so forth), while others are specific to the foreign exchange market (e.g., choice of numeraire, interest-rate adjustments for holidays, SDR-weighting market index and so forth). While some of the findings of Brown and Warner in the stock returns market also hold in this study, significant differences are found in many cases. For instance, Brown and Warner indicate that the mean-adjusted model performs well relative to the market model under a wide variety of conditions. In this study the performance of the market model usually dominates that of the mean-adjusted model unless the market index is also substantially affected by abnormal performance.

The results of this study provide some useful insights into the conduct of event studies in the foreign exchange market. The properties of sample-wide mean excess returns show that in some cases, the distributions are non-normal. Although the assumption of normality may be violated, the simulation results show that the *t*-test is generally robust to detect the absence or presence of abnormal performance. Nevertheless, the non-parametric Wilcoxon sign-rank test is slightly more sensitive than the *t*-test. In terms of currencies, special attention should be paid to the foreign currency. If the foreign currency is one of the EMS currencies (e.g., BF, DM, FF, NG), a .2% abnormal return level would be sufficient to be detected by the event study methodology. Alternatively, if the foreign currency is a free-floating currency (e.g., US, CD, JY, UK), a higher abnormal level of .3% to .5% is generally required.

Regarding the choice of the numeraire currency, the performance of the dollar is generally satisfactory. This is important to researchers since most currencies around the world are mainly traded against the dollar and many exchange rates are quoted in terms of dollars.

The results also show that a careful design of the methodology may improve its sensitivity. For instance, researchers should try harder to identify more relevant events to be included in the sample since sample size has significant impact on the sensitivity of the methodology. Nevertheless, a sample size of fifty is generally acceptable, while thirty may still be sufficient for the less volatile EMS currencies. Alternatively, it is not necessarily true that longer estimation periods provide better results. Frequent parameter shifts may render a longer estimation period less desirable. The optimal length varies with currencies and researchers need to refer to the findings in Table 5 before setting the estimation length for specific currencies. In addition, the simulation results show that it is generally acceptable to allow missing days (except days -1, 0) within the event period. Relaxing this restriction allows more eligible events to be included in the sample, thus increasing the power of the methodology.

Event studies in the stock market (e.g., Patell & Wolfson [1979]) find that substantial variance increases may occur around event dates. If this is true in the foreign exchange market, the results demonstrate that the sensitivity

of the methodology will be reduced. In general, both Type I & II errors will increase.

Event studies in the stock market generally do not adjust returns for holidays. However, in the foreign exchange market, estimation of holiday returns generally improves the power of the methodology. Furthermore, it is generally all right to neglect interest rates in the calculation of currency returns. This is useful news to researchers since event studies can now be applied to countries where data on daily interest rates are not readily available.

Regarding the choice of market index, Brown and Warner [1980] find that the equally weighted index may outperform the value-weighted index. This finding is confirmed in the foreign exchange market where the equally weighted index generally outperforms the GNP- and SDR-weighted indices. However, the difference is not substantial in some cases. Under these circumstances, the SDR-weighted index may be an acceptable market proxy. Since the SDR consists of only five commonly traded currencies, the computation of the market return can be simpler and the data easier to obtain.

In recent event studies in stock markets, researchers such as Linn and Pinegar [1988] tend to use a specific standardized abnormal returns procedure to derive test-statistics on abnormal returns. Although this approach reduces Type II error, it is generally accompanied by a moderate increase in the Type I error. This shortcoming is especially obvious when the foreign currency is the deutsche mark or Netherland guilder. Researchers therefore need to weigh the pros and cons to decide whether standardization procedures are appropriate for a particular event study.

As in Brown and Warner [1985], when the precise date of the event is not known, the power of the tests is reduced substantially. Type I and II errors are increased significantly. While a higher shock level of 1% may alleviate the problem for EMS currencies, this may still be inadequate for the free-floating currencies. It is therefore important to identify the precise date of the event if possible. For the convenience of the readers, the findings mentioned above are summarized in Table 9.

As in other studies, limitations are found in the present study. Unlike the stock market, where data are available for numerous different stocks over very long periods of time, the data in the foreign exchange market are more restricted. The reason that this study is confined to eight major currencies is that daily interest rates of other currencies are not available. It is understood that these eight currencies cannot exactly represent the entire market. Nevertheless, they represent the frequently traded currencies and constitute the major portion of the market. Additionally, because of the population from which samples are drawn, it is certain that a given event date may be reused a number of times, leading to cross-sectional dependencies among the draws in a given sample. We realize that we have reused our data set substantially in performing the simulations. With 50 draws per

TABLE 9
Summary of Simulation Results

Simulations	Remarks
Simulations 1: Varying Currencies	
1.1 Numeraire: US\$ and Foreign Currency: DM	Market model performs best, followed by market-adjusted, mean-adjusted, and random walk models in that order.
1.2 Numeraire: US\$, Foreign Currency varied	As foreign currencies, EMS currencies perform better than free-floating currencies. Market-adjusted model outperforms the mean-adjusted model except CD which is closely tied to US\$, the numeraire. The yen as the foreign currency has the lowest rejection rate.
1.3 Varying the Numeraire (56 tests)	US\$ is a fairly dependable numeraire. Relationship between the foreign currency and the numeraire is not necessarily reciprocal.
1.4 Sample of Foreign Currencies	Market model is still the best. The US\$ is the best numeraire among the eight currencies.
1.5 Market Index Affected by Shocks	If the effect is mild, the market model still dominates. When the weight factor increases to .5, the performance of the market-adjusted and market models is inferior to that of the random walk or mean-adjusted models.
Simulations 2: Varying Shock Levels, Time Periods and Variance	
2.1 Varying Shock Levels	At 0% shock level, Type I error is not substantial regardless of models used. A .2% shock seems to be adequate for most EMS currencies as foreign currencies. To reach satisfactory rejection rates, free-floating currencies require .3 to .5% shock.
2.2 Increase of Variance in Event Period	Both Type I and II errors are increased.

Simulations 3: Sample Size, Length of Estimation Period, Missing Days within Event Period	3.1 Varying Sample Size (30, 50 and 100)	Sample size matters. 50 seems to be an acceptable level while 100 is better. While 30 may still be all right for EMS currencies as foreign currencies, it may not be sufficient for free-floating currencies.
3.2 Allowing Missing Days on Non-critical Days		Generally, allowing missing days within the event period except days -1 and 0 improves the power of the methodology. The only obvious exception is when BF is the foreign currency. If missing days are not allowed, sample size may be seriously compromised.
3.3 Varying Length of Estimation Period		A longer estimation period may not necessarily produce better results. It varies over different foreign currencies and numeraires. Researchers need to check with information reported in Table 7.
Simulation 4: Interest-Rate Adjustments		Elimination of interest rates from currency return calculation is generally all right. Estimation of returns for non-trading days by an averaging procedure increases the power of the test.
Simulation 5: Choice of Market Return Proxy		Generally, the equally weighted index outperforms the GNP- or SDR-weighted indices. If there are data constraints, the SDR-weighted index is still acceptable except in the case when the foreign currency is the Canadian dollar.
Simulation 6: Standardization of Abnormal Returns		Standardization helps reduce the Type II error especially when the foreign currency is a free-floating currency. However, Type I error is also increased after standardization. The increase is especially obvious when DM or NG is the foreign currency.
Simulation 7: Imprecise Event Date		Type I and II errors increase substantially. A shock level of 1% may be required to attain satisfactory rejection rates when the foreign currency is an EMS currency. However, this shock level is still inadequate for the free-floating currencies. Knowing the precise event date is therefore important.

sample, each employing 61 days of data, 3050 days are drawn from the 2650 days of available data for one sample. This is repeated 100 times to perform one test between a numeraire and foreign currency. The constraints on available data forced this compromise of sound procedure. However, if one is willing to accept the view that the historical time-series of currency returns reasonably well proxy the time-series from which future event studies are executed, the findings of our study are transferable.

Finally, when interest-rate adjustments are made for the holiday returns, the returns are based on a geometric mean procedure. Such treatment violates the assumptions on which the statistical tests are based. Even though violations are committed, empirical findings show that the adjustment improves the results. The ad hoc adjustment method proves to be practical and useful in controlling both Type I and Type II errors.

It is hoped that the findings of this study provide useful information to researchers who intend to conduct event studies in foreign exchange markets. Since not all of the findings of Brown and Warner hold in the foreign exchange market, the authors would encourage similar simulation studies to be conducted in other financial markets (such as preferred stocks, bonds, options and commodities) in order that the reliability of event studies can be increased in those markets.

NOTES

1. The exchange rates are the averages of bid rates collected from five New York banks at noon. Alternatively, the interest rates are day-to-day domestic bid rates in various national markets. The European interest rates are collected off the screen around 11:00 a.m. Eastern time in the U.S. The U.S. interest rate is the effective Fed Funds rate. The interest-rate data are given on a per annum basis and need to be converted into daily rates before being used in calculating currency returns. Two bases are used in calculating interest rates for domestic deposits: 360-day year and 365-day year. Excepting Sterling, all the currencies included in this study use the 360-day year basis; interest in Sterling is calculated on a 365-day basis.
2. The currency return can be alternatively expressed in log form. Simulations were conducted to compare the results under the two different return specifications. The results were essentially the same. This paper therefore reports the results of the first form of return specification which is more frequently seen in the literature.
3. The authors would like to thank Rene Stulz for suggesting the addition of the random walk model to the list of models examined.
4. A second version of random walk model is also found in the foreign exchange market. Some foreign exchange forecasters believe that the best forecast for the future spot exchange rate is the present spot rate. Thus, the expected change of exchange rate is a martingale. Frenkel [1981] and Meese & Rogoff [1983] have found empirical support for this model. If this model is valid, the overall currency return will be a submartingale with expected value equal to the interest differential between the two currencies. Simulations were conducted to compare the results of these two random walk models. The results were very similar. This occurs since the daily interest differential (annual interest differential divided by 360) is too small to materially impact results. Only the results of the first random walk model will be reported.
5. In Brown & Warner [1980, 1985], no adjustment is made to the market return on the event date even when one of its securities is selected and shocked. This is understandable since the induced abnormal performance of one security is unlikely to change the market return which is an average return of thousands of securities. However, in the present event study of foreign exchange markets, a

selected foreign currency represents one-seventh of the market. When abnormal performance is introduced to the foreign currency return, the market return will also be affected. Two approaches are considered in this study to address this problem. The first approach is to leave the foreign currency out when the market return is computed on the event date. The second approach is to leave the foreign currency in with the added abnormal performance. Simulations were conducted and the results under these alternative approaches were found to be quite similar. Consequently, the standard experimental setup adopts the first approach which is more consistent with Brown & Warner [1980, 1985]. This issue will be discussed in greater detail under Simulation 1.5.

6. Only 1132 out of the original 2650 observations are represented here. For an observation to be eligible, data on exchange rates as well as interest rates must be available for all eight currencies on that day. This is required for the calculation of the market index.

7. Different simulations are conducted to examine the event study methodology in foreign exchange markets. Some of the simulations are adapted from Brown & Warner [1980, 1985] while others are specially prepared for the foreign exchange market (e.g., choice of numeraire). Issues that are not important in the foreign exchange market are left out though they may be important in the common stock market. For instance, the issue of non-synchronous trading is much discussed in Brown & Warner [1985]. Such issue is not important in the currency markets since most of the hard currencies are traded and quoted every day. It is true that the governments of less developed countries (LDCs) may sometimes suspend the trading of their currencies. Nevertheless, these LDC currencies are regularly managed by governments and should not be used for event study to begin with.

8. As discussed in Brown & Warner [1980, p. 216], the percentage of rejections is also a random variable with a sampling distribution. Assume that, under the null hypothesis, the outcomes of the hypothesis tests for each of the 100 replications are independent. By employing a Bernoulli process, rejections are expected 5% of the time with a standard deviation of .022 when testing at the .05 level. If the percentage of rejections is normally distributed, the percentage of rejections should be from 0.7% and 9.3% approximately 95% of the time in Table 2 when there is no abnormal performance on a given day. At the .01 level, rejections should be between 0% and 2.9% approximately 95% of the time.

In comparing the percentage of rejections across different models, a guide is needed in assessing if differences are statistically significant, either pairwise or jointly. If, for each of the 100 replications, the results for two different models are independent of each other, then the differences in proportions test is used to determine the standard deviation of the difference in rejection percentages (given 100 replications). This required difference can be as high as 7% when the rejection percentages are around 50% level. At the upper or lower ends when the rejection percentages are around 99% or 1%, the standard deviation is reduced to 1.4%.

9. The close linkage of U.S. and Canadian currencies, for example, is demonstrated by examining the correlation coefficient of exchange rates of .93408 and return rates of .93262 occurring when the Japanese yen is used as the numeraire. A discussion in Simulation 1.5 considers the circumstances and consequences of closely linked currencies when they are either strongly or weakly represented in the market index.

10. Nevertheless, it is possible to find cases such that only the selected EMS currency shows abnormal performance on the event date while the other EMS currencies remain undisturbed. Firstly, an EMS member currency is allowed to deviate 2.25% (6% for the Italian lira) from the central rate before member countries are required to take action in the foreign exchange market to force the rate back within the 2.25% trading band. This 2.25% is much larger than the .2% shock level which is enough to signal significant abnormal performance. Secondly, the definition of currency returns includes an interest-rate component. Some of the reaction to the event can be captured by interest-rate changes in the following days; it is unnecessary to have all the adjustment made in the exchange rates.

In fact, intervention begins when a member's currency deviated more than 1.69% (4.5% for the Italian lira) from its central rate, reaching the "threshold of divergence." Even so, the 1.69% is still much larger than the .2% shock level which is already adequate to show significance differences under the event study methodology.

11. Alternative simulations were conducted in which even the foreign currency is given a shock level randomly selected from a uniform distribution (0% to .4%). The findings are similar and the same conclusions can be drawn.

12. A simulation similar to that of 1.4 was also conducted where the foreign currency was randomly drawn and the market index was affected by the shock. Similar conclusions were drawn. The performance of the market and market-adjusted models deteriorates when other currencies in the market index are also affected by the shock.

13. Note 8 provides information on assessing if the differences among models and estimation periods are significant.

14. Alternatively, simulations were conducted to include the return of the foreign currency in the calculation of the market return under the three weighting systems. Instead of six currencies, the market basket consisted of seven currencies. Similar results were obtained.

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